

GOKULRAJ NATARAJAN

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PROFESSIONAL SUMMARY

Forward-thinking Electronics and Communication Engineering undergraduate at **SIMATS University, Chennai** (Class of 2029) maintaining an **83.8% academic score (8.38 CGPA)**. Hands-on builder passionate about embedded hardware, microcontroller firmware (C/C++), and real-time IoT architectures. Proven track record in designing deterministic sensor nodes, failsafe power-switching circuitry, and winning 1st Place / Best Capstone Project.

TECHNICAL SKILLS

Hardware & Microcontrollers: ESP32, ESP8266, Arduino (Uno/Nano/Mega), Raspberry Pi, STM32 basics, Circuit Design, Breadboard Prototyping, MOSFET/Relay Drivers.

Sensors & Peripherals: MPU6050 (6-DOF IMU), NEO-6M GPS, SIM800L GSM, Ultrasonic (HC-SR04), IR Sensor Arrays, TDS & Fluid Sensors, LDR Photocell, Bluetooth (HC-05/BLE), Rotary Encoders.

Firmware & Programming: Embedded C, C, C++, Python 3, Interrupt Service Routines (ISR), State Machines (FSM).

Protocols & Cloud: I2C, SPI, UART, PWM, Wi-Fi, HTTP/MQTT, Blynk IoT Cloud, Bluetooth SPP Serial, USB HID.

Tools & Simulation: Arduino IDE, VS Code, Git, GitHub, Proteus, EasyEDA, Oscilloscope Debugging, Multimeter Diagnostics.

Core Engineering: Sensor Calibration, Signal Filtering, Anti-Bounce State Lock, Real-Time Hardware Telemetry.

EDUCATION

Bachelor of Engineering in Electronics and Communication Engineering (B.E. ECE) 2025 — 2029 (Expected)
SIMATS Engineering (Saveetha Institute of Medical and Technical Sciences), Chennai • Academic Score: 83.8% (8.38 CGPA)

FEATURED HARDWARE & IOT PROJECTS

IoT Enabled Automated Precision Dosing System

2026

ESP32 / MCU, Power MOSFET Switching, Fluid Sensor Matrix, Blynk Cloud Telemetry

- Engineered an automated closed-loop fluid dosing platform integrating microcontrollers and power-switching circuits to regulate chemical reagent injection with $\pm 1.5\%$ precision.
- Developed remote monitoring telemetry for real-time stroke-rate modulation, tank level tracking, and automated emergency shut-off thresholds.

IoT Smart Parking Management Infrastructure **AWARD: BEST CAPSTONE PROJECT**

2025

Multiplexed IR Beam Arrays, Ultrasonic Transducers, Low-Power Interrupt Wakeups

- Architected a scalable smart parking system utilizing sensor grids for sub-second bay occupancy detection, reducing parking search latency by 40%.
- Awarded **1st Place / Best Capstone Project Award** at SIMATS University for superior circuit layout, energy-efficient idle standby (~40% power draw), and end-to-end reliability.

Elder Guard — Wearable Fall Detection & SOS Handshake

2026

MPU6050 6-DOF IMU, NEO-6M GPS, SIM800L GSM, Custom Jerk-Threshold Filters

- Built a wearable emergency safety tracker integrating 6-axis accelerometer/gyroscope with GPS geofencing and instant cellular alert dispatch.
- Implemented dynamic jerk-threshold math filters achieving >95% fall classification accuracy, coupled with a novel phone-ring handshake mechanism to cancel false triggers.

Smart Classroom Occupancy & Energy Automation

2026

Directional IR Gate Arrays, LDR Photocell, Bluetooth Serial SPP, Power Relays

- Built an automated room monitoring node using directional IR sensors with custom state-lock debounce algorithms to eliminate sensor bounce during crowded student transits.
- Automated lighting and ventilation control based on real-time headcount telemetry, delivering ~30% idle power savings.

Custom Automotive Steering Wheel & Low-Latency HID Interface

2026

High-Res Rotary Encoders, Tactile Switch Matrix, USB Serial HID Firmware

- Designed an ergonomic, modular steering controller with custom tactile inputs and interrupt-driven firmware for ultra-low latency (<5ms) in simulation and telemetry tracking.

CERTIFICATIONS & HONORS

- 1st Place / Best Capstone Project Award:** Winner for IoT Smart Parking Management System (SIMATS University, 2025).

- **NPTEL Certification:** Internet of Things (IoT) — In-depth protocols, cloud integration, and sensor networks.
- **Cisco Networking Academy:** Computer Networking Fundamentals & Network Architecture Certification.
- **The Bharat Scouts and Guides:** Certified for Leadership, Discipline, and Team Collaboration.